Biochemistry Cume
November 10, 2006

I. The venom of the stiletto snake (*Atractaspis bibroni*) contains a peptide toxin that is a potent vasoconstrictor. It exhibits significant sequence similarity to human endothelin 1. The primary amino-acid sequence is shown below:

cys-ser-cys-ala-asp-met-thr-asp-lys-glu-cys-leu-tyr-phe-cys-his-gln-asp-val-ile-trp

A. (10 points) Write out the 1 letter code for the above peptide.

B. (20 points) Calculate the net charge of the peptide at pH 3 and at pH 7.

C. (20 points) Write out the products you would generate if you treated a portion of the peptide separately with trypsin, chymotrypsin and cyanogen bromide. Write out the products for each. Use the one letter code.

D. (10 points) The chymotrypsin treated toxin was applied on a DEAE column. Write out the order of emergence from the column.

E. (10 points) When treated with trypsin, the protein migrates on polyacrylamide gels under nonreducing conditions with the same molecular weight as the native protein. However, prior denaturation with urea + beta-mercaptoethanol, followed by removal of the beta-mercaptoethanol by dialysis, produces lower molecular weight peptide(s) upon trypsin treatment. What does this information tell you?

II. (20 points) Draw a di-peptide Ala-Pro in cis-conformation at pH 7. Draw all of the hydrogen atoms.

III. Glycophorins are red blood cell membrane glycoproteins that bear important blood-group antigens. Glycophorin B possesses a single transmembrane segment. The primary sequence of glycophorin B (Blanchard et al., J. Biol. Chem. 262 (1987) 5808) is:

Leu Ser Thr Thr Glu Val Ala Met His Thr Ser Thr Ser Ser Ser Val Thr Lys Ser Tyr

Ile Ser Ser Gln Thr Asn Gly Glu Met Gly Gln Leu Val His Arg Phe Thr Val Pro Ala

Pro Val Val Ile Ile Leu Ile Ile Leu Ser Val Met Ala Gly Ile Ile Gly Thr Ile Leu

Leu Ile Ser Tyr Ser Ile Arg Arg Leu Ile Lys

(10 points) Identify the transmembrane segment.